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Managing adaptive performance in teams: Guiding principles and behavioral markers for measurement

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ABSTRACT

Various types of organizations must manage rapidly changing operational contexts. To respond to these demands, organizations are relying more heavily on team-based work arrangements. Effectively managing such performance requires a systematic, broad approach to measuring team effectiveness that is comprehensive and sound, yet unobtrusive. One aspect of teamwork that is critical to success in these types of environments is adaptation. Teams must be able to react quickly and accurate to the changing environment. To effectively manage adaptive team performance in such contexts, there is a need to better understand team adaptation as well as to generate better team performance measurement systems. To this end, a review and synthesis of the empirical, theoretical, and methodological literatures concerning team adaptation, performance, and measurement is conducted to develop theoretically-based principles to guide development of effective team adaptation measurement systems as well as to inform future research. We propose six guiding principles that capture core features of team adaptation and serve as an aid in the development of team performance measurement systems. These principles are rooted in recent theoretical work on team adaptation and are presented at a level of abstraction suitable for generalization across performance measurement contexts and purposes. Behavioral markers describing processes associated with each principle and example measurement strategies are presented to illustrate development of specific measurement tools and metrics, based on the principles. The principles and behavioral markers presented can guide development of measurement systems to assess, train, and improve team adaptation, a core capacity of effective organizations. Future research needs to expand upon the principles advanced here to provide theoretically grounded and methodologically rigorous tools to help performance management professionals develop adaptive team capacities.

Nearly four decades ago, Terreberry (1968) argued that adaptability would become crucially important to organizational success due to externally-induced organizational change and those organizations successful at adapting would be most effective in the marketplace. This forecast could not have been more prescient. Organizations increasingly rely on teams to manage the complexity of modern work. This continuing shift from organizing units of work around individuals to teams is being driven, in part, by factors such as the rate of change in task demands and the pervasive nature of change in business environments (Kozlowski, Gully, Nason, & Smith, 1999). Additionally, the same factors spurring creation of team-based organizations impose demands on teams, once formed, to rapidly and continuously adapt their performance processes.

To address this emerging need, the available theoretical and empirical literature on team adaptation has increased significantly in recent years (e.g., Burke, Stagl, Salas, Pierce, & Kendall, 2006; Chen, Thomas, & Wallace, 2005; Deshon, Kozlowski, Schmidt,

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Milner, & Wiechmann, 2004; Entin & Serfaty, 1999; Kozlowski et al., 1999; Lepine, 2003, 2005). Throughout the scientific literature, researchers describe team adaptation as a complex phenomenon, one comprising multiple inputs, interaction processes, and emergent states that results in event-driven changes in team properties and processes, enabling higher levels of effectiveness in complex environments.

Managing the adaptive performance of teams is, therefore, of great practical concern to organizations as well as a fertile ground for scientific investigation. However, much of the relevant scientific literature has yet to be integrated into the processes by which organizations manage performance. This article addresses this gap by providing an initial synthesis of theory and methods from the science of teams. This effort is intended to provide guidance on measuring and managing adaptive team performance and organizing future research.

Specifically, this article addresses three primary goals. First, the theoretical drivers of team adaptation are reviewed and summarized in an extended framework of processes and emergent states. This framework outlines the content a team adaptation measurement system should capture. Second, six guiding principles for measuring team adaptation are presented. These principles are rooted in the theoretical review as well as the team performance measurement literature. Third, example measurement strategies based upon each guiding principle are presented to illustrate the process of implementing the principles presented in this article. Before addressing these goals, the need for including the measurement of team adaptation in performance management systems is discussed.

1. The need for measuring team adaptation in team performance measurement systems

In a 'matrixed organization' where project teams are formed to address specific and unique client needs from pools of personnel with different types of expertise, the effectiveness of a team is in large part a function of the quality of the inputs to this process such as the characteristics of the individuals selected for the team (Bell, 2007). An extensive body of work is available to guide organizations in managing the performance of the individuals in this scenario. However, team interaction processes as a whole also contribute greatly to the effectiveness of outcomes (LePine, Piccolo, Jackson, Mathieu, & Saul, 2008; DeChurch & Haas, 2008). Consequently, a comprehensive approach to performance management in situations such as the one described above must represent aspects of both the individual and the collective. Sole focus on one level at the expense of the other can lead to performance decrements (e.g., DeShon et al., 2004) and maladaptive performance such as social loafing or lack of focus on collective goals.

Aguinis (2009) outlined six purposes of a performance management system: strategic, administrative, communication, developmental, organizational maintenance, and documentation. In the context of adaptive team performance, the strategic (i.e., linking goals across levels), communication (i.e., providing personnel with information on their level of performance), and developmental (i.e., providing feedback to drive coaching) are the most salient goals of a performance management system. Measurement underlies the capacity to *systematically* manage performance and meet each of these goals. However, performance management research and practice has not widely incorporated the concepts or methods of measuring team adaptation. The remainder of this article presents the state of the science of team adaptation and its measurement to both inform practice and guide future research into this growing research and application domain.

2. Theoretical drivers

In this section, a review of the theoretical basis of team adaptation and related concepts is provided in order to outline the content of a team adaptation measurement system. Theories of team adaptation serve as the primary theoretical driver of the principles offered in this article; however, the general scientific literature on team performance processes, and team learning are reviewed as well. Each of these key theoretical drivers is discussed briefly below. Subsequently, an expanded conceptual framework of team adaptation based on these three areas is presented.

2.1. Team adaptation

There are several models of team adaptation and related phenomenon in the literature (e.g., DeShon et al., 2004; Kozlowski et al., 1999). Most recently, a cross-level mixed determinants model was developed by Burke et al. (2006). Integrating several perspectives from the organizational, behavioral, and cognitive sciences, this multidisciplinary, multilevel and multiphasic model provides an account of the processes involved in team adaptation. Burke et al. (2006) propose that the *process* of adaptive team performance, defined as "an emergent phenomenon which compiles over time from the unfolding of a recursive cycle… to functionally change current cognitive or behavioral goal directed action" (p.1192), is an antecedent to the *outcome* of team adaptation. Team adaptation is conceptualized as "a change in team performance, in response to a salient cue or cue stream that leads to a functional outcome for the entire team" (p.1190). To reach this outcome, the team engages in a cycle of adaptive team performance which, as shown in Fig. 1, consists of four process-oriented phases: (1) situation assessment, (2) plan formulation, (3) plan execution, and (4) team learning. In situation assessment, the team gathers and interprets information. This information is used to create a course of action in plan formulation, which is then set in motion during plan execution. Once the plan has been executed, the team reflects on past events and learns from their experiences. These processes lead to corresponding emergent cognitive and affective states (i.e. shared mental models, team situational awareness, psychological safety) which function as both

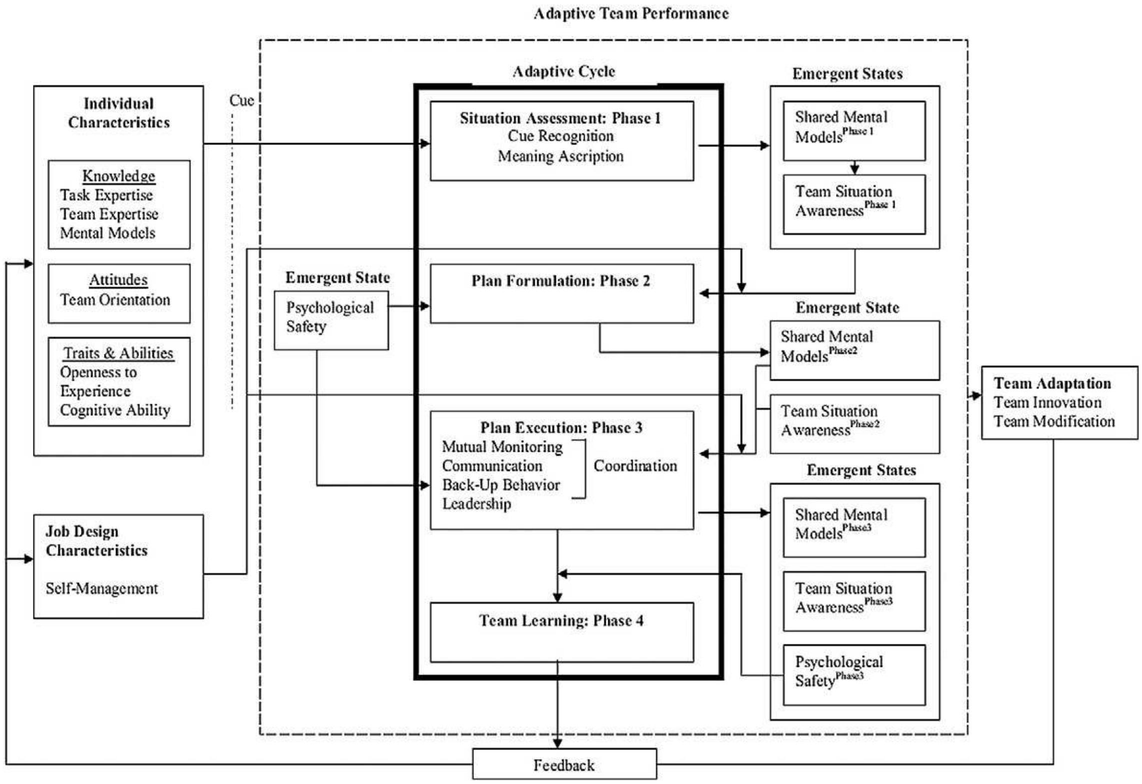

Fig. 1. Input-throughput-output model of team adaptation (from Burke et al., 2006).

proximal outcomes and inputs throughout the cycle. Adaptive team performance comprises these team processes and emergent states.

The Burke et al. (2006) model of team adaptation provides a parsimonious, approach to conceptualizing team adaptation. It serves as a useful point of departure for the present purposes because of its dynamic and multi-level nature. The following sections expand on this model to provide a more detailed picture of processes and emergent states involved in teamwork.

3. Team performance processes

The model developed by Burke and colleagues provides an account of the means by which a team can change its performance processes (i.e., how the team adapts); however, for the sake of parsimony, it does not include a detailed account of the variety of performance processes (i.e., cognitive or behavioral goal directed action) that may be acted upon by adaptation processes. Based on the assumption that different team processes are critical at different phases of the team performance cycle, Marks et al. (2001) developed a recurring phase framework and taxonomy of team processes that has been validated via meta-analysis (LePine et al., 2008). This framework and the literature it is based upon serve as a second theoretical driver. Through a review of existing taxonomies of team processes (e.g., Fleishman & Zaccaro, 1992; Prince & Salas, 1993), Marks and colleagues compiled a comprehensive list of ten dimensions critical to team functioning. Transition processes, such as mission analysis, goal specification, and strategy formulation occur during planning activities. Action processes, such as coordination, monitoring, and back-up behavior occur during the periods of time when teams conduct activities leading directly to goal accomplishment. Interpersonal processes – conflict management and affect management – occur in both transition and action phases. Applying this temporal framework to the Burke et al. (2006) model, plan formulation consists of transition processes, and plan execution of action processes, with interpersonal processes occurring in both.

3.1. Team learning

Team learning is the last phase of the Burke et al. (2006) adaptive cycle and consequently, the extant theory on team learning serves as a third theoretical driver. Edmondson (1999) argued that team learning occurs by members testing assumptions and discussing differences openly. This process begins with information sharing and review of past events. Kasl, Marsick, and Dechant (1997) proposed that the processes involved in team learning can be divided into two component parts (thinking, action) and integration. In the thinking component, team members reflect on past performance, interpret the consequences of their actions, and diagnose strengths and weaknesses. In the integration component, the team summarizes the lessons learned and develops a plan for integrating these lessons into the team's processes in future performance episodes.

In sum, the Burke et al. (2006) framework of team adaptation serves as the fundamental basis for the principles developed in the following section. This model details the processes by which a team alters its internal performance processes and resulting emergent states in response to salient cues. In addition to this central theoretical driver, the principles presented draw upon existing literature on team performance processes in general as well as team learning. Specifically, the work of Marks and colleagues provides a complimentary view of team performance over time and helps to add a more detailed view of the full range of processes a team engages in during a performance episode. Additionally, the available literature on team learning provides a deeper insight into the processes involved in the final phase of team adaptation. In the following section, we detail an expanded framework of team adaptation that includes a more detailed description of the team processes discussed above.

3.2. Expanded framework of team adaptation

The model proposed by Burke et al. (2006) provides a parsimonious model of the core phenomenon of interest. However, as outlined above, team adaptation involves a larger set of teamwork processes and emergent states. In this section, we provide a more comprehensive and detailed framework of these processes and emergent states. This framework more fully maps the construct space a team adaptation measurement system should consider. The expanded framework is more comprehensive and corresponding processes within the original model have been updated based on a recent review of team process literature. The framework still consists of four phases within the adaptive cycle: situation assessment, plan formulation, plan execution, and team learning. Detailed descriptions of the phases as well as the sub-processes involved in each phase follow.

3.2.1. Situation assessment

Situation assessment refers to an individual level process of information gathering in which at least one team member scans the environment for cues that have the potential to affect the success of the mission (Burke et al., 2006; Gutwin & Greenberg, 2004). This is a recurring process which exerts temporal influences on the individual actions of team members throughout the life cycle of the team. Team members essentially are individually searching for potential problems that the team might encounter and trying to make sense of the implications of these problems and generate initial solutions. This process is critical to the team's success — specifically, research has noted that the speed at which (1) environmental cues are identified and (2) responses are generated is related to how successful the team is at adaption (Smith-Jentsch, Johnson, & Payne, 1998; Waller, 1999).

There are several key sub-processes within the situation assessment phase, including cue recognition and meaning ascription. *Cue recognition* refers to the identification of cues or cue patterns, focusing on issues that might negatively impact or have already negatively impacted the mission success (Burke et al., 2006; Salas, Rosen, Burke, Goodwin, & Fiore, 2006). Fleishman and Zaccaro (1992) suggest that acquiring information is a leadership activity — a process of compiling and storing raw information related to the mission. This is similar to cue recognition within our framework; however, we do not limit this task to the leader. Any team member can acquire relevant information.

Once identified, cues must be translated into information that is meaningful to the team. *Meaning ascription* refers to the process of assigning meaning and relevance to cues by classifying or synthesizing them based on existing knowledge (Burke et al., 2006; Cannon-Bowers, Tannenbaum, Salas, & Volpe, 1995). In other words, raw information is not sufficient for triggering team adaptation. Acquired information must be given meaning before the team can decide how to respond to it. For example, if a team of training designers is doing an audience analysis for their intended training, they may determine that their audience is primarily senior-level executives. This information may be interpreted as meaning the training should skip past basic material and focus on higher-level content, or as an indication that the training should employ more traditional methodology since the executives are less technology savvy than younger, new employees. Either way, these interpretations can be classified as meaning ascription.

3.2.2. Plan formulation

In plan formulation, teams generate a sequence of actions designed to transform the current or expected state of the environment into the desired or goal state (Burke et al., 2006). This involves the simultaneous tasks of planning and problem solving, two distinct activities (Klein & Miller, 1999). For example, planning does not necessarily involve problem solving when addressing routine issues since experienced individual team members may know exactly how to handle a situation they have previously encountered. Alternatively, problem solving does not always result in a plan (i.e., a sequence of actions). Regardless of the degree to which teams engage in planning and/or problem solving, plan formation is a critical aspect of team performance in both relatively static environments, where a plan can be fully developed and implemented as designed, as well as in more dynamic environments, which require continual planning in order to encompass the changes in the environment. In those dynamic cases, plans function as a guiding framework, aiding team members with their interpretation and reaction to the environment in a dynamic yet coherent manner (Mathieu & Schulze, 2006). In the framework presented here, plan formulation is comprised of the following processes: mission analysis, goal specification, deliberate planning, contingency planning, role differentiation, and preemptive conflict management. Each of these processes is discussed in more detail below.

Mission analysis is the multilevel process of interpreting and evaluating the characteristics of a team's mission including tasks, subtasks, and available resources (Marks et al., 2001; Prince & Salas, 1993). Individual and teams engage in analytical reasoning and intuitive pattern-recognition based diagnosis of the situation/problem (Prince & Salas, 1993). These are cognitively rooted processes at the individual level as a team member works to understand the problem and resources available. These individual level representations are subsequently communicated to the team and can potentially be altered based on input from other team members (Cooke, Salas, Kiekel, & Bell, 2004).

Goal specification is the explicit identification, articulation, and prioritization of the team's goals in the context of the mission. Clear goal specification includes setting expectations about (1) what must be accomplished, (2) within what time frame and (3) to what degree of quality (Marks et al., 2001). Both mission analysis and goal specification are multilevel processes that seek to move the situational understanding or awareness of the individuals comprising the team into a structured and defined team understanding. Both of these processes are crucial to problem solving as the manner in which a problem is conceptualized or represented drives much of the solution generation process (Hayes, 1989; Klein, 1998).

After the problem has been defined, the team can begin *developing a course of action* to address the problem, assuming that the team members do not have a priori access to a plan that satisfactorily addresses the situation at hand. If this is available to the team, they will quickly move toward later stages of the adaptive process such as plan execution, described more fully below. However, developing a plan involves strategy formulation wherein the team proposes and develops multiple courses of action intended to accomplish the previously specified goals given the constraints identified during mission analysis. Strategy formulation can be classified into three sub-dimensions: deliberate planning, contingency planning, and reactive strategy planning (the latter is discussed in plan execution). *Deliberate planning* is the “formulation and transmission of a principal course of action for mission accomplishment” (Marks et al., 2001, p. 365), which basically refers to the development and communication of ‘Plan A,’ (i.e. the sequence of actions developed based on the best information and reasoning available to the team at the time). However, critical pieces of information may be missing or, in more dynamic environments, information critical to mission success can change after the team has engaged in deliberate planning. Therefore, teams also engage in *contingency planning*, which is the “a priori formulation and transmission of alternative plans and strategy adjustments in response to anticipated changes in the performance environment” (Marks et al., 2001, p. 366).

Role differentiation occurs throughout plan formation. Team members set responsibility boundaries within the context of the mission/task (Kozlowski et al., 1999). This involves critically examining member resources, skills, abilities, and prior knowledge, and subsequently matching them to subtask requirements (Fleishman & Zaccaro, 1992; Kozlowski et al., 1999; Sutton, Pierce, Burke, & Salas, 2006).

Preemptive conflict management refers to the deliberate process of delineating guidelines to aid in the prevention and control of team conflict prior to the occurrence of conflict (De Dreu & Weingart, 2003; DeChurch & Marks, 2001; Marks et al., 2001). The purpose of this process is twofold: (1) develop a shared understanding of how the team feels about conflict and (2) determine how and when team members will address conflict. The establishment of norms for cooperative approaches to conflict resolution help avoid competitive or aggressive tactics.

3.2.3. Plan execution

Plan Execution has been defined as “an assortment of concomitant individual- and team level processes that are enacted dynamically, simultaneously, and recursively” (Burke et al., 2006, p. 1195). This constitutes the performing phase of team adaptation: team members actively engage in many affective, behavioral, and cognitive processes in order to successfully execute the plan outlined during plan formulation. These include coordination, mutual monitoring, back-up behavior, systems monitoring, reactive strategy planning, reactive conflict management, and affect management.

At the core of plan execution is *coordination*, described as the orchestration of the sequence and timing of interdependent actions (Marks et al., 2001). Coordination involves pacing activities within determined temporal boundaries (Fleishman & Zaccaro, 1992; Kozlowski & Ilgen, 2006; Salas, Sims, & Burke, 2005). Coordination is a team level process, and can take place explicitly, implicitly, or in some combination of the two (Burke et al., 2006; Entin & Serfaty, 1999). Various processes affect plan execution both directly and indirectly by helping to enable effective coordination which, in turn, allows for the successful timing and execution of the plan.

Mutual monitoring is a cognitive process in which team members observe their fellow team members' actions and behaviors in order to catch and correct performance discrepancies. Formally defined, it is the “team members' ability to keep track of fellow team members' work while carrying out their own… to ensure that everything is running as expected and… to ensure that they are following procedures correctly” (McIntyre & Salas, 1995, p. 23). Mutual monitoring facilitates coordination of actions through the perception and mitigation of performance discrepancies.

Team members participating in mutual monitoring perceive when other members need help and subsequently engage in assisting behaviors known as *Back-up Behavior* (Porter, Hollenbeck, Ilgen, Ellis & West, 2003; Salas, Priest & Burke, 2005). Back-up Behavior can take many forms such as providing a teammate verbal feedback, assisting a teammate behaviorally in carrying out actions, or completing a task for a teammate (Marks et al., 2001). If a team recognizes that a certain team member is overloaded, work can be shifted to underutilized team members in order to balance the workload.

Another type of monitoring behavior that takes place during plan execution is *systems monitoring*. Systems monitoring is the tracking of team resources and environment conditions as they relate to mission accomplishment (Marks et al., 2001). Systems monitoring behaviors can be classified into two categories: (1) internal systems monitoring and (2) environmental monitoring. Internal systems monitoring involves tracking resources internal to the team such as personnel, equipment, and information generated by or contained by the team. Environmental monitoring focuses on conditions external to the team that could affect plan execution. Systems monitoring differs from mutual monitoring in that resources are tracked rather than team member behavior.

Sometimes a plan runs into unanticipated changes in the environment, and revisions must be made mid-execution to the existing strategy. This is known as *reactive strategy planning* (Marks et al., 2001), and it is the adjustment of strategies based on information gathered from the environment and performance feedback. This form of planning falls into the plan execution phase rather than the planning phase because it is essentially an adaptive redirection of the team's activity. This type of adaptable behavior has been supported as a critical mechanism of teamwork (Salas, Sims et al., 2005).

The presence of conflict can have many negative effects on team performance. It can disrupt the coordination of activities, cause a lack of mutual monitoring and back-up behavior, inhibit communication, and consequently bring about a loss of trust and motivation. Thus, it is imperative to minimize conflict during plan execution. The first strategy is to prevent conflict (see preemptive conflict management above). Once it has occurred, however, *reactive conflict management* is the process of working through task, process, and interpersonal disagreements among team members (Marks et al., 2001). Management of conflict during plan execution must occur as quickly as possible in order to reduce the amount of disruption to other processes. Reactive conflict management can involve many strategies, including negotiation and mediation.

Affect management is a concept that is closely related to conflict management. Just as conflict can lead to detrimental affects on team performance, negative affectivity can also lead to negative outcomes. Heightened emotional states, such as frustration and anger, can lead to poor performance, and can also be psychologically unhealthy for the individuals. Many factors can inflate or deflate emotional levels during plan execution such as hostility between members, frustrating task conditions, or failures in execution. Strategies such as team building activities, stress-relieve training or even joking and complaining, when used properly, can be effective for affect management.

3.2.4. Team learning

Team learning can be thought of as a continuous learning process whereby permanent changes in the team's behavioral repertoire emerge as a result of the acquisition, combination, and sharing of new knowledge (Edmondson, 1999; Kasl et al., 1997). This is in large part a deliberative process where a team retrospectively evaluates its past performance with the goal of changing its processes to produce higher levels of performance outcomes. As indicated in Fig. 2, there are four categories of processes related to team learning: *recap* (wherein the team builds understanding of its past performance), *reflection* (wherein the team diagnoses and evaluates its performance), *integration* (wherein the team transforms the initial understanding of the team's performance into a new, shared model by incorporating previously identified successes and failures), and *action planning* (wherein the team purposefully develops lessons learned from the evaluation of its past performance and develops a plan for integrating these lessons into the team).

Recap involves the team building an understanding of what happened during past performance episodes. Team members search and structure information on an individual level by perceiving, filtering, and storing cues related to task performance (Burke et al., 2006; Kraiger, Ford, & Salas, 1993; van Offenbeek, 2001). Additionally, teams share their own thoughts as well as seek information and feedback from their teammates for a more robust understanding of both their own individual and team level

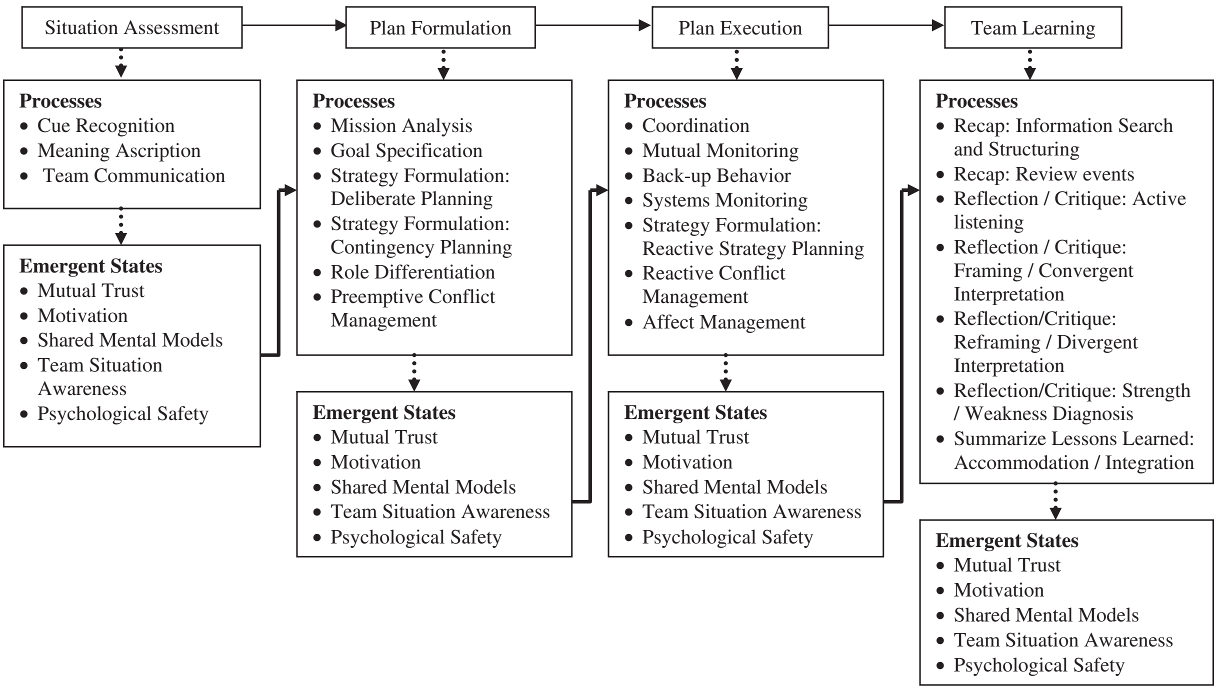

Fig. 2. Expanded framework of team adaptation.

performance (Smith-Jentsch, Zeisig, Acton, & McPherson, 1998). Individually, team members analyze prior adaptability cycles for relevant information. As a group, members should discuss noted errors or near misses, which serve as diagnostic signs of larger issues (Ilgen, Hollenbeck, Johnson, & Jundt, 2005). Much of this discussion can be done through storytelling, which helps facilitate situational awareness as well as accommodate the need of team members to voice a range of beliefs (Burke, Salas & DiazGranados, 2008).

Reflection focuses on turning the shared understanding of the past performance episode into a functional improvement for the team by analyzing which performance strategies were successful and which need improvement (Edmondson, 1999). The team transforms the recap of events into a coherent picture of the team's performance processes, outlining errors and successes, often referred to as framing/convergent interpretation (Kasl et al., 1997; van Offenbeek, 2001). Subsequently, effective teams engage in reframing and divergent interpretation of their understanding of past performance episodes by sharing differing opinions about causes of effective and ineffective performance and engaging in 'what if' scenarios to assess past performance (Kasl et al., 1997; van Offenbeek, 2001).

During *integration*, team members build a coherent picture of team performance processes by integrating previously identified successes and failures (Kasl et al., 1997; van Offenbeek, 2001). Successful integration occurs when the team identifies points of agreement and disagreement about the causal mechanisms underlying successes and failures. Teams discuss points of disagreement until consensus and common understanding is achieved. As noted above, integration should focus on both successes and areas requiring improvement, which requires a more granular focus on specific processes, rather than just overall performance.

Finally, in the *action planning process*, previous discussions are integrated and formalized into plans designed to improve future team performance (Smith-Jentsch et al.,1998). By summarizing lessons learned from past performance, teams can see areas requiring improvement. Members should discuss how a course of action can be revised to allow for more a more seamless and fluid performance episode.

3.2.5. Emergent states

Emergent states are defined as "properties of the team that are typically dynamic in nature and vary as a function of team context, inputs, processes, and outcomes" (Marks et al., 2001, p. 2001). These cognitive, motivational, and affective states are produced as team members interact during a performance episode. Emergent states have been conceptualized as either moderators of team performance processes (e.g., Ilgen et al., 2005) or as proximal inputs and outputs for team processes (i.e., team interaction produces an emergent state which subsequently influences future team processes; Burke et al., 2006). Below we provide a description of the emergent states included within this model of team adaptation.

Mutual trust is an emergent affective state critical to team effectiveness (Salas, Sims et al., 2005). As an individual attitude, trust is a belief or expectation that another individual will behave in a manner that is indicative of good intentions (Spector & Jones, 2004). Therefore, it follows that *mutual* trust is a "shared belief that team members will perform their roles and protect the interests of their teammates" (Salas, Sims et al., 2005, p. 561). Each phase has differential impacts on the development of mutual trust, and in turn, trust influences each subsequent phase. For example, the role differentiation process within plan formation is critical to the development of trust. This is due to the fact that one integral component of trust is the belief that members will perform their assigned tasks. Mutual trust developed from the plan formulation phase will, in part, determine the willingness of other members to engage in mutual monitoring, which requires a willingness to communicate.

Motivation is the result of a process of allocating energy to different actions with the goal of achieving the greatest satisfaction of needs (Naylor, Pritchard, & Ilgen, 1980; Pritchard & Ashwood, in press). Individual behavior is a result of the direction, intensity and persistence of effort (Pritchard & Ashwood, in press). Within the team adaptation framework as illustrated in Fig. 1, motivation influences each phase. Individual team members each have a certain degree of motivation at the inception of the team, which is then affected by the various team adaptation processes throughout the duration of the team. When generalizing individual motivation to a team level construct, it is important to consider the implications and, more importantly for the purpose of this discussion, the influence one has on the other. When applying social learning theory principles (Bandura, 1997) to the concept of motivation, it can be thought of as contagious: individuals who are highly motivated to do their work may influence other team members' attitudes (Chen, Kirkman, Kanfer, Allen, & Rosen, 2007) — reflective of an individual influence over the team. Conversely, individual members may feel more competent to perform their tasks if the rest of the team has high competency beliefs as well (Chen et al., 2007) — team influence over the individual. A key implication is that motivation will exert influence over individual and team level behaviors throughout the process and both individual and team level motivation can influence adaptive team performance.

Shared mental models and *team situation awareness* are two additional emergent states that serve as proximal inputs and outcomes throughout all four phases of team adaptation. Shared mental models can be conceptualized as dynamic, simplified, cognitive representations of reality used to describe, explain, and predict events (Burke et al., 2006; Langan-Fox, Code & Langfield-Smith, 2000). Team situation awareness refers to the shared understanding of the current situation at any given point in time (Salas et al., 1995). Situation awareness refers specifically to the team's understanding of their current situational surroundings, while shared mental models can refer to any aspect of reality that is understood by the team. These emergent states serve as a cognitive framework that allows the team to predict future system states. Without an accurate shared understanding of the current situation, teams cannot adapt.

Finally, each phase influences *psychological safety* as well. Psychological safety, defined as the shared belief that the team is "a safe place" for interpersonal risk taking (Edmondson, 1999). A climate of psychological safety enables team members to speak up and contribute during plan formulation. Psychological safety, while important in all phases, has particular weight in several processes within this model. For example, during plan development, if members feel like contributions will be valued, the team can generate more ideas during brainstorming.

Fig. 2 summarizes the above discussion and provides an expanded framework of the team performance processes involved in adaptive team performance over time. This framework sacrifices the parsimony provided by the Burke model for inclusiveness. This inclusiveness is necessary for driving measurement systems capable of providing a robust and diagnostic profile of changes in team performance overtime (see Salas, Rosen, Burke, Nicholson & Howse, 2007). In the following section, we provide six guiding principles based upon the framework detailed in Fig. 2 and the underlying theory discussed above.

4. Measuring team adaptation: principles and behavioral markers

In this section, we present six guiding principles for developing measurement systems capable of capturing the complexity of team adaptation. First, each principle is introduced and described. Second, each principle is discussed in relation to the expanded framework of processes involved in adaptive team performance over time to illustrate the components of team adaptation each principle is designed to capture. Third, examples of measurement strategies based on each principle are provided. Measuring team adaptation requires an approach akin to triangulation where multiple measures captured through multiple methods and observers is necessary to capture the complexity of the phenomenon. Consequently, the example strategies pull from several methodological approaches. Table 1 provides a review of the principles as well as an overview of some of the core measurement methods in team research that map to each principle. These methods will be referred to throughout the following sections.

As summarized in Table 1, behavioral markers have been drawn from the theoretical and empirical literature on the processes and emergent states in the expanded team adaptation framework. These behavioral markers are statements that serve as indicators or the presence or absence of the associated construct and can be used as a starting point for developing content for the various measurement strategies. In addition to theory, the purpose of measurement (e.g., assessment, research, feedback and remediation during training) is a primary driver of the development of a team performance measurement system (Brannick & Prince, 1997). While the principles presented in this section are at a level of abstraction applicable across purposes and contexts, the manifestation of each principle in any one measurement system may vary depending on these factors. Example measurement strategies are discussed for a range of purposes and contexts, and suggestions are given regarding how these example measurement tools could be used to manage adaptive team performance in organizations.

4.1. Principle #1: capture bottom-up changes in team performance

An effective team adaptation metric must capture changes in the patterns of unique lower-level contributions of team members. Team adaptation occurs in dynamic environments. Without change, adaptation is unnecessary. In rapidly changing environments, teams must alter performance processes 'on the fly,' in response to unanticipated task demands. This involves shifting workload and task responsibilities between team members to ensure that the team's goals are met. In this way, team adaptation can be viewed as a change in the pattern of performance contributions of individual team members in response to an unforeseen change or novel situation (Kozlowski et al., 1999; LePine, 2005). If a measurement system is not sensitive to shifts in the pattern of performance changes across team members (e.g., by collapsing individual performance into aggregate team performance), a fundamental aspect of team adaptation will be absent from the measurements.

Principle #1 deals primarily with the plan execution phase of team adaptation. Plan execution, as defined previously, at team level is the performing phase of team performance: team members actively engage in many cognitive and behavioral processes, and a measurement tool must capture changes in these processes. As shown in Fig. 2, several processes within plan execution involve changes in lower-level contributions of team members: *coordination, back-up behavior, reactive conflict management,* and *affect management.* Some example behavioral markers from these processes are included in Table 2.

4.1.1. Example measurement strategies

There are a number of methodological options available for capturing the components of team adaptation associated with Principle #1. As this principle primarily relates to the plan execution phase of team adaptation, the large literature base concerning measuring the performance of action/performing teams (Sundstrom, De Meuse, & Futrell, 1990) is relevant (e.g. Brannick, Salas, & Prince, 1997; O'Neil & Andrews, 2000). This includes such methods as the analysis of communication content and flow, observer ratings, and self-assessments of performance. For example, Behaviorally Anchored Rating Scales (BARS) and Behavioral Observation Scales (BOS), can be used to capture the processes involved in plan execution (for a review of these and other fundamental methods of team performance measurement, see Kendall & Salas, 2004; Salas, Priest, et al., 2005). Essentially, the example markers in Table 2 can be used to generate specific rating forms to capture characteristics and frequencies of team processes such as coordination and back-up behavior.

While the development of observational ratings scales is appropriate for measuring aspects of adaptation in teams performing in a naturalistic setting, event-based measurement offers unique opportunities for measuring team adaptation in controlled simulations. Event-based measurement (Fowlkes, Dwyer, Oser, & Salas, 1998) focuses on systematically identifying and creating measurement opportunities by embedding trigger events into simulation scenarios. This technique is commonly used to diagnose team performance during training (Fowlkes & Burke, 2006; Fowlkes, Lane, Salas, Franz, & Oser, 1994); however, by controlling

Table 1
Behavioral markers of team adaptability.

Team adaptability processes	Markers	Selected references
Coordination	• Articulate information about their status, needs and objectives as often as necessary (and not more) • Sequence team task work behaviors to minimize downtime within interdependent tasks • Synchronize teamwork behaviors without overt communication in high workload conditions • Pass information to one another relevant to the task in a timely and efficient manner • Resolve any conflicting task demands through role negotiation	• Kozlowski et al., 1999; Fleishman & Zaccaro, 1992; Marks et al., 2001
Back-up behavior	• Proactively assist fellow team members with task work • Provide feedback to fellow team members to facilitate self-correction • Redistribute workload and adjust resources to fit task requirements across all roles in the network, i.e. workload balancing	Kozlowski et al., 1999; Fleishman & Zaccaro, 1992; Marks et al., 2001
Reactive conflict management	• Identify the parameters of conflict between team members • Utilize negotiation or mediation strategies for conflict resolution	Marks et al., (2001)
Affect management	• Identify problematic emotions that negatively impact team • Regulate problematic emotional responses • Provide opportunities for development of social cohesion	• Marks et al., (2001)
Mission analysis	• Explicitly articulate the team's objectives • Discuss the purpose of the team in the context of the present performance environment • Discuss how the available team resources can be applied to the meeting the team goals • Agree on the methods and approaches the team should take to work towards its goal (e.g., low task conflict related to selection of performance strategies)	• Marks et al., (2001)
Goal specification	• Identify what needs to be accomplished, within what specified time frame at what level of quality • Explicitly define desired outcomes • Prioritize the required actions of the team for mission success	Marks et al., (2001)
Strategy formulation: deliberate planning	• Explicitly articulate expectations for how a proposed course of action should unfold • Brainstorm plans and strategies for solving problems • Categorize, condense, combine and refine expectations into mutually agreed upon list of additional tasks and information requirements	• Marks et al., (2001)
Strategy formulation: contingency planning	• Explicitly articulate expectations for how an alternative course of action • Brainstorm plans and strategies for solving problems • Categorize, condense, combine and refine expectations into mutually agreed upon list of additional tasks and information requirements	• Marks et al., (2001)
Strategy formulation: reactive strategy planning	• Modify routine performance strategies in response to unexpected environmental or task changes	Marks et al., (2001)
Role differentiation	• Match member KSAs to subtask requirements • Explicitly state boundaries of responsibilities between team members	• Sutton et al., 2006; Kozlowski et al., 1999; Fleishman & Zaccaro, 1992
Cue recognition	• Scan environment for cues that might influence the outcome of the mission • Rapidly detect problems or potential problems in their environment	• Burke et al., 2006; Stagl, Burke, Salas, & Pierce, 2006
Meaning ascription	• Explicitly define potential problems • Accurately assess the underlying causes of environmental changes • Categorize issues based on relevance to mission success • Map relationships between relevant information and mission success by identifying impact of cues on mission • Generate solutions to potential problems by drawing upon mental models of previously used courses of action	• Stagl et al., 2006
Team communication	• Clearly communicate problem definitions • Follow up to ensure that messages are received and understood • Acknowledge messages when they are sent • Cross check information with the sender, to ensure that the message meaning is understood • Articulate 'big picture' to one another as appropriate • Proactively pass information without being asked	Smith-Jentsch, Johnson et al., 1998; Stagl et al., 2006; Fleishman & Zaccaro, 1992
Mutual monitoring	• Observe team members' performance to identify errors • Identify unbalanced workload distributions	McIntyre & Salas, 1995; Kozlowski et al., 1999
Systems monitoring	• Track internal systems resources such as personnel, equipment, and other information that is generated or contained with the team (p. 367) • Track the environment conditions relevant to the mission (p. 367)	Marks et al., 2001; Fleishman & Zaccaro, 1992

(continued on next page)

Table 1 (*continued*)

Team adaptability processes	Markers	Selected references
Recap: information search and structuring	• Filter and store the filtered information concerning past performance. • Perceive cues and corresponding information concerning past performance • Seek information and feedback from other team members about past performance.	Mohammed & Dumville, 2001
Recap: review events	• Refer to past adaptation/performance cycles. • Summarize key features of past performance episode.	• Ilgen et al. (2005)
Reflection/critique: active listening	• Accept reasonable suggestions. • Respond to other's requests for information. • Avoid sarcastic and non-constructive comments.	• Edmondson, 1999; Ilgen et al., 2005
Reflection/critique: framing/convergent interpretation	• Frame the situation to provide the initial perception of the past performance episode. • Discuss errors.	• Kasl et al., 1997; van Offenbeek, 2001; Mohammed & Dumville, 2001
Reflection/critique: reframing/divergent interpretation	• Integrate and share views of causes of effective and ineffective past performance. • Engage in 'what if' scenarios to reinterpret causes of past performance.	Kasl et al., 1997; van Offenbeek, 2001
Reflection/critique: strength / weakness diagnosis	• Discuss the consequences of their actions. • Discuss how unintended consequences can be avoided. • Discuss how an ineffective course of action or process can be revised.	Smith-Jentsch, Johnson et al., 1998; Mohammed & Dumville, 2001
Summarize lessons Learned: Accommodation/integration	• Articulate plans for changes in future performance processes. • Team members create new mental models by integrating viewpoints. • Team members decide and act on new mental models.	Ilgen et al. (2005)

the nature of the events and the number of events through the use of scripts, 'opportunities to adapt' can be engineered and consequently, measurement can be tied to these opportunities. For example, back-up behavior can be assessed using this approach by creating circumstances in a simulation that produce unbalanced workload and assessing the presence or absence of targeted responses based upon the behavioral markers provided in Table 2. Performance in these event-based simulations could be used to track the performance of teams in back-up behavior, and then to link the performance of those teams to individual and organizational goals as well as provide developmental feedback to the teams on how they can improve their back-up behaviors. Fig. 3 illustrates how the behavioral markers for coordination can be used to generate a Behaviorally Anchored Rating Scale (BARS) item.

4.2. Principle #2: capture top-down changes in strategy that impact team performance

In addition to changes in individual member contributions that happen in the absence of a centralized or explicit shift in procedure, an effective team adaptation metric must capture strategy shifts within the team. In addition to the more reactive changes in performance patterns described above, adaptation can also occur in a more feed forward manner where teams adopt entirely new plans of action or make large scale modifications to their plans. In this regard, team adaptation can be viewed as a shift in the overall structure of the team in anticipation of changes in the task demands or environment. Therefore, a measurement system must not only capture the bottom-up changes in behavioral performance patterns described in Principle #1, but also the top-down changes in overall plans that are responsive to change in the environment or task. Major shifts in strategy tend to occur most often during the plan formulation phase of the team adaptation cycle comprising the following processes: *mission analysis,*

Table 2
Summary table of principles.

Principle	Example measurement approaches
1. Capture bottom-up changes in team performance.	• Communication content and flow analysis • Event-based measurement
2. Capture top-down changes in strategy that impact team performance.	• Observation scales • SME review of strategy plan
3. Capture a team's recognition of a need for change.	• Event-based measurement • Communication coding
4. Capture the team's ability to self-assess.	• Communication analysis • Behavioral observation ratings
5. Capture what team members are thinking and feeling in a dynamic manner	• Cue strategy associations • Card sorts • Questionnaires
6. Capture a profile of team adaptation over time	• Time-sensitive measurement

Coordination: the orchestration of the sequence and timing of interdependent actions (Marks et al., 2001)

1	………..	All team members sequenced their task work so that downtime was minimized on interdependent tasks.
2		
3	………..	Some (approx. 50%) team members sequenced their task work to minimize downtime on interdependent tasks.
4		
5	………..	None of the team members attempted to sequence their task work in order to minimize downtime on interdependent tasks.

Fig. 3. Example of a Behaviorally Anchored Rating Scale (BARS) item measuring coordination.

goal specification, strategy formulation, and *role differentiation.* Example behavioral markers from these processes are listed in Table 2.

4.2.1. Example measurement strategies

Several of the measurement approaches discussed above are useful for capturing aspects of team planning, the focus of Principle #2. Table 2 lists example behavioral markers for the processes involved in planning. These can be developed into observational scales and used in coding and rating communication and performance. For example, Behavioral Observation Scale (BOS) items can be developed for goal specification by further developing the markers into concise questions for which observers provide estimations of frequency. An example of such an item is provided in Fig. 4. Additionally, the artifacts or outputs of the processes discussed above can be captured and assessed. For example, a team's efforts at mission analysis can assed via questionnaire (e.g., team situational awareness; Salas et al., 1995), or communication content analysis (Cooke et al., 2004). Similarly, the formal plan or strategy developed by a team can be captured (e.g., through communication analysis in naturalistic settings or by requiring formal specification of a plan in more controlled experimentation). This plan can be evaluated by subject matter expert review or by comparison to an optimal solution if available. By capturing a formal representation of the team's plan or strategy, measurement of aspects of Principle #1 will be facilitated as well (i.e., knowing what the team's initial plan is will help identify intentional and adaptive deviations from that plan during plan execution). A team's plan at time two could be compared to their plan at time one in order to determine if a shift in strategy occurred, and the team could be given coaching regarding how their plan should have changed if it did not, or how it could be improved if it did.

4.3. Principle #3: capture the team's recognition of a need for change

An effective team adaptation metric must capture the team's sensitivity to cues that signal a need to change. In order for a team to effectively adapt to an environment, its members must be aware of a change in the environment or a decrease in the team's performance. A measurement system must capture team members' sensitivity to cues that signal a need for a change in performance processes and be diagnostic of strategy shifts and changes in patterns of performance contributions. This principle does not apply neatly to one phase of the model, but tends to encompass processes across phases including the situation

Question: Did the team prioritize the required actions of the team for its success?

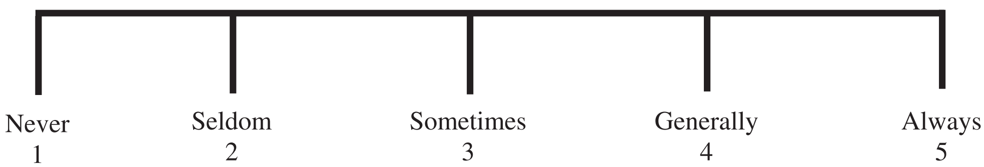

Fig. 4. Example of a Behavioral Observation Scale (BOS) item tapping goal specification.

assessment and plan execution phases. Situation assessment is an individual-level process of gathering information whereby at least one team member scans the environment for cues that have the potential to affect the success of the mission (Endsley, 1995; Gutwin & Greenberg, 2004), and it recurs throughout the duration of the team adaptation cycle. On the team level, this individual level dynamic understanding of the state of the environment is translated into a shared understanding within the team through communication processes (Salas et al., 1995; Cooke et al., 2004). Situation assessment includes the sub-processes of *cue recognition, meaning ascription,* and *team communication*. In the plan execution phase, similar processes take the form of *mutual monitoring* and *systems monitoring*. Markers of these processes are shown in Table 2.

4.3.1. Example measurement strategies

Principle #3 is designed to capture aspects of adaptive team performance that occur across several phases of the adaptive cycle. Because Principle #3 involves the team's ability to detect cues in the environment, event-based measurement is a valuable tool when it is feasible (i.e., when team performance is being measured in an environment where some control over environmental cues is possible). By purposefully manipulating cues in the environment, the precision of measurement can be increased as there is knowledge about when critical cues in the environment are present; however, this approach is not practical in many naturalistic environments. In situations such as these, observational tools and communication coding schemes can be developed from example behavioral markers in Table 2 corresponding to the processes subsumed under Principle #3. For example, characteristics of team communication have been associated with building and maintaining shared situational understanding (Cooke et al., 2004; Salas et al., 1995) and hence can be considered general or distal indicators of the team's ability to detect and critical changes in the environment. Additionally, detailed communication analysis can be used to track the flow of information between team members and provide developmental feedback regarding missing communication links or critical missing information. From this analysis, latency measures can be extracted (e.g., the amount of time taken for a critical piece of information to move from the team member that identifies it to the team member that needs it). Fig. 5 provides an example of an Event-Based Checklist item that captures aspects of cue recognition, communication, and meaning ascription.

4.4. Principle #4: capture the team's ability to self-assess

An effective team adaptation metric must capture the ‘selection’ of team performance processes. Adaptation involves a change in the performance processes of a team, but, of course, this is not change for the sake of change. The new performance processes must be functional for the team, meaning that they result in 1) higher levels of performance outcomes under the same task constraints, or 2) a recovery from decreased levels (or the potential for decreased levels) of performance outcomes caused by a disruption or shift in the environment or task. For a team to consistently exhibit adaptive behavior there must be a ‘selection’ process capable of culling ineffective performance processes from effective performance processes at the team level. Measurement must be sensitive to the deliberative processes the team engages in to diagnose its performance and alter future performance processes based on evaluations of past performance episodes.

4.4.1. Example measurement strategies

This measurement principle largely focuses on the team learning phase of the team adaptability framework. A large part of team learning involves looking back over the teams' past performance episodes to determine effective and ineffective performance processes. Many of the previously mentioned methods (e.g., observer ratings, communication analysis) are appropriate for capturing aspects of team learning. Several processes lend themselves to an observational scale, such as reviewing events and several of the reflection/critique processes as these involve observable behaviors like discussing errors, verbally summarizing past performance, and avoiding sarcastic and non-constructive comments. Similar to earlier examples, either the frequency (BOS) or the quality (BARS) of the behavior could be measured by constructing an appropriate scale from example behavioral markers included in Table 2 and observing team performance. However other processes within the team learning phase are more cognitive in nature and thus are not easily measured by observation (e.g., filtering and storing information). Self-report measures are better

Event	Critical Response	Hit
The team losses a direct communication channel with a distributed member.	Team members detect that communication has been lost *(cue recognition)*.	
	Team members communicate nature of the problem to the rest of the team *(communication)*.	
	Team members accurately assess the underlying cause of the down communication line *(meaning ascription)*.	

Fig. 5. Example event-based checklist item capturing aspects of cue recognition, communication, and meaning ascription.

suited to capture member activities that are more covert in nature. Self-report scales of team learning behaviors have been developed as well (e.g., Edmondson, 1999). A communication content analysis would be appropriate for certain research questions involving team learning processes such as discussing particular issues relevant to the team's progress.

4.5. Principle #5: capture what team members are thinking and feeling in a dynamic manner

An effective team adaptation metric must capture emergent cognitive and affective states. These emergent states can be viewed as holistic team properties that arise from the interaction of team member performance processes and the contextual factors of the environment in which the team is performing (Cooke et al., 2004; Cooke, Salas, Cannon-Bowers, & Stout, 2000). There a variety of methods available to measure and index static team knowledge (e.g., shared mental models; Cooke et al., 2000; Langan-Fox et al., 2000; Mohammed & Dumville, 2001), and while this is type of knowledge is a critical component of team effectiveness, understanding team adaptation requires a finer temporal resolution of measurement. Team members develop a rapidly changing awareness of the situation that is shared or distributed across team members (Artman, 2000; Gorman, Cooke, & Winner, 2006) especially in highly dynamic environments. Additionally, team level affect has been identified as critical to performance outcomes (Jordan & Troth, 2004; Kelly & Barsade, 2001), but most frequently it is treated as a more stable trait than a dynamic construct. This same problem has been noted with the conceptualization and measurement of affect on the individual level as well (Beal, Weiss, Barros, & MacDermid, 2005).

4.5.1. Example measurement strategies

Principle #5 involves capturing emergent states, both affective and cognitive. While techniques exist for accomplishing this, Principle #5 is possibly the most difficult principle to implement. There are several methods available for assessing emergent cognitive states such as shared mental models and team situation awareness (Cooke et al., 2000; Mohammed & Dumville, 2001). For example, the content of shared mental models can be assessed through various knowledge elicitation techniques (e.g., cue-strategy associations, declarative knowledge tests) and aggregated (for a review, see Cooke et al., 2000; Smith-Jentsch, 2008). Similarly, the structure of shared mental models can be assessed as well through techniques such as analysis of paired comparison ratings, or card sorts (Cooke, 1994). Developmentally, the team could discuss their mental model sharedness and determine ways to improve their shared understanding of their work in order to improve performance. Strategically, assessments of mental models could be used to determine whether or not the teams have a shared strategic vision, and whether or not that strategic vision follows that of the organization.

Emergent affective states are most frequently assessed via self-report methods. Questionnaires are available to assess trust, motivation, and psychological safety. However, there is a need for better measurement approaches in this area. A recent study found that paired comparisons were the best measurement technique for assessing quality and similarity of strategic consensus mental models (Resick et al., in press). Yet administration of questionnaires, for the measurement of both cognitive and affective states, is obtrusive, making it difficult to obtain a fine grained temporal resolution with currently existing measurement techniques (i.e., questionnaire-based measures frequently cannot be administered repeatedly). More research and development of methods and tools for dynamic and unobtrusive ways for measuring these constructs is needed.

4.6. Principle #6: capture a profile of team adaptation over time

An effective team adaptation metric must capture performance changes over time. Team adaptation is a process that unfolds over time as the team repeatedly engages in performance episodes, discovers new means of accomplishing tasks, and responds to changes in the task environment. Capturing a single snapshot of team performance is inadequate to assess team adaptation. The preceding five principles are designed to guide the development of measurement systems so that the fundamental processes involved in team adaptation are captured. However, this measurement must be extended over time to yield an accurate understanding of how a team responds to changes. Therefore to adequately diagnose the effectiveness of team adaptation over time, measurement must 1) be based in the five principles outlined above and extended over time, and 2) capture team level performance and significant changes in the task or environment. Measuring adaptation fundamentally involves measuring change which has been a long-standing and fundamental issue in the social and behavioral sciences (Cronbach & Furby, 1970). However, statistical techniques such as latent growth modeling, multiple indicator latent growth modeling, mixed effects modeling (Bliese & Ployhart, 2002; Chan, 2002), and other advances in time series analysis (Shumway & Stoffer, 2006) offer an increasingly robust tool set for researchers to analyze how teams adapt performance processes over time.

4.6.1. Example measurement strategies

Implementing Principle #6 involves the collection and analysis of multilevel longitudinal data. Fundamentally, team adaptation is an outcome, a functional outcome, meaning that change resulting from the adaptive team performance cycle must contribute to the development or maintenance of the team as a unit. So, in addition to issues discussed in Principles #1 through 5 (which focus on the team processes of adaptive performance), the functional outcomes of the team must be tracked over time (Burke et al., 2006). By plotting team performance trajectories over time, a nonlinear pattern should emerge where satisfactory levels of performance are punctuated by transitions periods (i.e., declines in performance that indicate a misalignment of performance processes followed by realignment and increased performance levels; Chen & Ployhart, 2004).

5. Concluding remarks

The goal of this article is to address a critical issue in performance management: a focus on the effective measurement of team adaptation. To this end, the authors integrated literature on team performance, adaptation, and measurement in development of six principles to guide efforts regarding team adaptation measurement. The six principles are descriptive of the core features of team adaptation. These theoretically-grounded principles are derived from an understanding of performance measurement systems that address teamwork in particular. Specifically, such systems must capture both changes in the patterns of lower level performance contributions of team members and more global shifts in performance strategies in response to changes. This represents capturing aspects of the 'bottom up' and 'top down' nature of adaptive team performance.

Fundamentally, adaptation is about response to change. Thus, comprehensive measurement systems in team-based environments must capture the processes by which a team senses external changes in the environment or tasks. Additionally, these measurement systems must capture the processes by which a team deliberately analyzes past experiences and codifies these experiences into changes in team performance processes during future performance episodes. As emergent cognitive and affective states are tightly coupled with team processes and team context, they too are a critical component to a comprehensive team-based measurement system that includes adaptive performance. Lastly, managers should not settle for snapshots of performance. To capture team adaptation, the preceding principles must be applied to investigate performance changes *over time*. These principles and example behavioral markers have been developed to help bridge the gap between recent theory on team adaptation and the need for more robust approaches to measuring, and ultimately managing, team adaptation within modern organizations.

The guiding principles and example measurement strategies are offered to address needs identified by Salas and colleagues (2009), specifically, needs for better theories of team adaptation and better measurement practices in team research. A comprehensive performance management system must capture all important aspects of performance to avoid deficiency. Given the strategic importance of teams in most organizational contexts, and the dynamic nature of business work processes, team adaptation measurement is of critical importance to managers focused on continuous improvement through systematic performance measurement. We hope this article encourages dialog about specific measurement metrics and tools aimed at capturing the dynamic process of team adaptation, and provides guidance for practitioners and academics seeking to capture the critical aspects of team adaptation to improve overall performance of both the teams, as well as the organizations as a whole.

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